A Spatiotemporal Fusion Transformer Model for Chlorophyll-a Concentrations Prediction Over Large Areas With Satellite Time Series Data

Gaoxiang Zhou, Ming Liu, and Liangzhi Li

***Abstract*— Predicting Chlorophyll-a (Chla) is essential to support the marine environment changes and marine ecosystem health, and provide early warning of algae blooms. The development of learning-based methods has facilitated Chla prediction research. Still, most of the current methods can only predict short-term Chla changes in small areas, which is limited by the ability of the model to exploit spatiotemporal dependencies. Thus, this article proposes a spatiotemporal fusion transformer prediction model (STF_Transformer) to predict relatively long-term Chla changes (15 days ahead). This model utilizes temporal and spatial transformer modules to extract the temporal and spatial correlations of the input spatiotemporal sequences, which are then fused to predict the 15-day Chla. The experimental results show that the proposed model has the optimal performance compared to the existing methods [e.g., convolutional neural network (CNN), long- and short-term memory (LSTM), and convolutional LSTM (ConvLSTM)], with root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) less than 0.61 mg/m^3, 0.315 mg/m^3, and 22.5%, respectively, for 15 days Chla prediction in a large area (including Bohai, Yellow, and East China Sea). In addition, the temporal and spatial prediction results of the proposed model show that the predicted Chla has consistent temporal and spatial patterns with the observed Chla. This study indicates that the proposed STF_Transformer model can provide a highly accurate prediction of Chla over a large area in the relatively long term (15 days), providing data and technical support for marine ecosystem-related applications.**

***Index Terms*— Chlorophyll-a (Chla) prediction, large areas, satellite, spatiotemporal, transformer.**

I. Introduction

CHLOROPHYLL-A (Chla) serves as a crucial indicator of changes in marine ecosystems [1]. It is an intuitive characterization of phytoplankton biomass and plays a key role in studying ocean dynamics processes and the impacts of human activities and climate change on the marine environment [2], [3]. Thus, Chla prediction can provide important data and technical support in the study of marine environmental change [4], ecological health [5], ocean currents [6], and fishery management [7]. It is also a prerequisite for reducing the adverse effects of harmful algal bloom disasters and facilitating emergency response [8].

The study of Chla prediction can be divided into two categories: physical-based methods and data-driven methods. Physics-based methods mainly utilize ecological dynamics and flow-diffusion theory for Chla prediction [9], as demonstrated by a model in [10] that predicts ocean chlorophyll at various time scales (e.g., seasonal and perennial) and a biogeochemical model in [11] that predicts global ocean chlorophyll up to nine months ahead. Although physics-based methods can dynamically demonstrate various ocean physical processes and obtain predictions at different scales, these methods usually require the precise input of a large number of initial parameters, which is difficult to access in practice [12], [13], [14]. In addition, the accumulation of prediction errors in models can limit their ability to make robust long-term predictions in practical situations [15].

With the accumulation of satellite remote sensing data, data-driven methods, which can be categorized as traditional and deep learning (DL)-based approaches, have received remarkable attention in Chla prediction [16], [17], [18]. Generally, the data-driven approach relies on observational data to predict ocean variables and does not require prior knowledge and ocean physical processes. It learns potential evolutionary patterns by mining spatial and temporal variations of ocean variables in historical data [13]. Initially, traditional learning-based methods, such as random forest, artificial neural networks, and ensemble learning, have been widely introduced and driven by satellite data for the Chla prediction [19], [20], [21]. For instance, [22] integrated various wavelet nonlinear networks for achieving Chla prediction. However, these traditional learning-based methods are usually suitable for dealing with relatively simple nonlinear correlations and unsuitable for predicting spatiotemporal evolutions in the ocean [23].

DL-based methods are capable of extracting nonlinear relationships with fewer predetermined assumptions, and thus show promising performance in estimating and predicting marine environmental parameters [24], [25]. To date, multiple

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DL networks have been utilized in this field, such as recurrent neural network (RNN) [25], convolutional neural network (CNN) [26], long- and short-term memory (LSTM) [27], [28], CNN-LSTM [29], and convolutional LSTM (ConvLSTM) [17]. Specifically, the gate mechanism of LSTM can solve the problem of modeling the temporal dependencies of the input Chla sequence. CNN, on the other hand, extracts the spatial dependencies around the target pixel by convolutional kernel. Overall, the CNN-based method aims to learn spatial correlations of images, while an RNN/LSTM method focuses on learning temporal correlations between input sequences [18]. These characteristics make them suitable for short-term predictions of regions with less variability. However, learning only temporal or spatial features will limit the performance of a predictive model because Chla is inherently spatially and temporally correlated, especially at large scales over the medium to long term.

Consequently, spatiotemporal learning methods (ST_LMs) have been developed, such as CNN-LSTM, ConvLSTM, and modified ST_LM [12], [13], to improve the extraction of spatial and temporal features in Chla prediction. For example, the CNN-LSTM model combines spatial features captured by the CNN with temporal patterns identified by the LSTM, resulting in better prediction accuracy than using either model individually [29]. The ConvLSTM-based model integrates convolutional operation into LSTM, which can simultaneously extract spatiotemporal features of input spatiotemporal sequence data for prediction. However, ConvLSTM suffers from the loss of long-term spatiotemporal features, which limits the ability of medium- and long-term prediction [30]. Thus, researchers in [12] constructed a spatial attention and a temporal attention extraction module for obtaining spatial variability and temporal dependency features of Chla predictions, respectively. Additionally, a recent study proposed a spatiotemporal model based on graph neural networks (GCNs) for predicting ocean environmental variables (sea surface temperature and Chla). This model consists of spatiotemporal features, bias calibration, and periodic-dependent extraction modules, and achieved good results on monthly composite data [13]. Although these methods improve the capability of spatiotemporal feature extraction, there are still some issues, such as insufficient long-term dependency extraction by LSTM and the global spatial feature extraction restriction of CNN/GCN. In addition, most of the current DL-based Chla prediction studies focus on short-term prediction (less than or equal to nine prediction steps) in small regions (e.g., the Bohai Sea) [12], [17], and the prediction accuracy decreases obviously with the increase of prediction steps. This is also probably attributed to the insufficient ability of the prediction models to extract long-term spatiotemporal dependencies.

Multiple attention mechanisms, capturing explicitly or implicitly long- or short-term spatiotemporal dependencies, have recently been widely utilized for spatiotemporal prediction [31], [32], [33]. In this mechanism, each element in the time series will compute attention with all other elements, thus retaining and propagating the important information in the long sequence [34]. Building on the initial transformer, multiple transformer variants have been developed for applications in different domains (e.g., air, weather, and video prediction) to achieve high-precision long-term or short-term predictions [35], [36], [37], [38]. These studies typically segment spatiotemporal sequence data along the spatial dimension and then construct new patch sequences, which are fed into the transformer framework for prediction [32], [38], [39]. Given that Chla time-series data exhibit inherently spatial and temporal correlations, developing a transformer-based model that captures these dependencies from both temporal and spatial perspectives could enable more accurate Chla predictions.

To address this, our study proposed a spatiotemporal feature fusion prediction model based on a transformer framework (STF_Tranformer) to simultaneously predict medium-term (15-day) Chla across a large region. In this model, a temporal transformer module is used to extract the long/short-term dependencies, and a spatial transformer module is used to extract the global spatial dependencies in the spatiotemporal image sequences. The training and validation data are obtained from satellite-derived Chla daily products from 2012 to 2022. Comparison experiments with satellite observations are conducted to evaluate the model performance. This study will help quantify in advance large-scale changes in the marine environment, as well as study the response of marine ecosystems to natural and human activities.

II. Methodology

A. Problem Formulation

Considering the effect of historical spatiotemporal sequence evolution on future Chla change, the Chla prediction in this study can be described as the following problem: given the historical Chla image sequences $S^{\text{chla}}_{1:T} = \{C_1, \ldots, C_T\}$, where $S^{\text{chla}}_{1:T} \in \mathbb{R}^{T\times N\times N}$, $C_T \in \mathbb{R}^{N\times N}$ is derived from satellite remote sensing with image size $N \times N$ in the tth time node. The objective is to predict $S^{\text{chla}}_{T+1:T+L}$ as follows:

$$\hat{s}^{\text{chla}}_{T+1:T+L} = \text{argmax}_{S^{\text{chla}}_{T+1:T+L}} \text{P}\left(S^{\text{chla}}_{T+1:T+L} | S^{\text{chla}}_{1:T}\right) \tag{1}$$

where $\hat{s}^{\text{chla}}_{T+1:T+L} = (\hat{C}_{T+1}, \ldots, \hat{C}_{T+L})$ are the predicted Chla sequence at the center of the input image from $T+1$ to $T+L$.

B. Spatiotemporal Feature Fusion Transformer Model for Chla Prediction

Fig. 1 demonstrates the architecture of the proposed spatiotemporal fusion transformer prediction model (STF_Transformer) model based on the transformer framework. The input of the STF_Transformer is the historical Chla data $S_{1:T}$ and output is the future L-steps (L is 15 days in this study) Chla data $S_{T+1:T+L}$ at the center of the input image. STF_Transformer is established based on the transformer encoder framework and consists of two submodules: the temporal transformer module (TTM) and the spatial transformer module (STM). Both modules aim at extracting potential spatiotemporal dependencies in spatiotemporal sequence data. The inputs are flattened from a temporal perspective (or spatial perspective) into a vector (of length T or $N \times N$) for different submodules. Each element of the vector is projected into the embedding dimension by a linear transformation, and then positional embeddings

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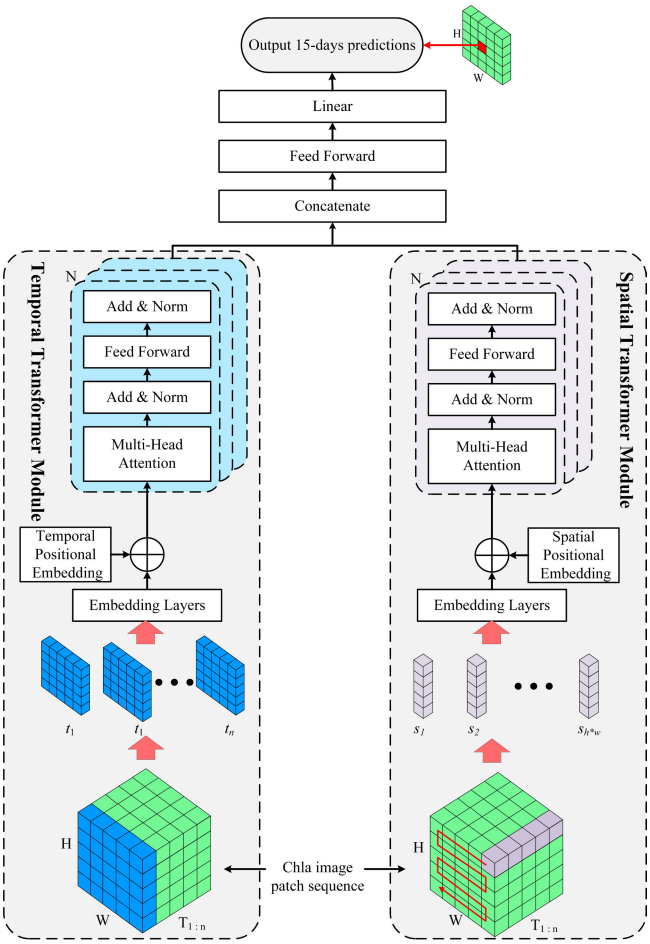


Fig. 1. Flowchart of the proposed STF_Transformer.

are added to generate the final input. The structure of the two submodules is similar to the transformer encoder, but enhances the extraction of temporal and spatial features of image time series in terms of input data reorganization and positional embedding, respectively. The outputs of submodules are concatenated to generate a spatiotemporal coded representation. The feedforward layer is then used to reduce the length of the coded representation vector. Finally, the output of the feedforward layer is fed into the linear regression layer to obtain the final prediction. Note that the number of layers of the encoder is set to three.

1) Transformer Encoder: The transformer encoder, operated in parallel with high efficiency [34], was utilized as a basic framework to develop a prediction model from the historical Chla spatiotemporal time series. Typically, the transformer encoder is constructed from multihead attention layers, feedforward layers, and normalization layers. For brevity of the article, the multihead attention, as the core of the transformer encoder, is introduced as follows and the details of other components and the original transformer can be referred to [34] and [36].

First, an embedding layer is used to map each element of raw input sequence X into a space of higher dimensions, each of which can represent a different feature. Taking the Chla time series as an example, the original 3-D data $X_e \in \mathbb{R}^{t\times w\times h}$ can be converted from the time dimension to $X_e \in \mathbb{R}^{t\times(w\times h)}$, where t is the length of the time series, and $(h \times w)$ is the unfolded data for each time stamp. The converted data is then input into the embedding layer to get a new feature $X_e \in \mathbb{R}^{t\times d}$, where d is the size of the embedding dimension.

Then, X_e will be partitioned along the dimensions of the embedding space, thus generating a vector for each head with dimension $(l, d/h)$. Then attention is obtained from Values (V), Key (K), and Query (Q) results, which are obtained by the linear transformation layer. The most common attention function is the scaled dot product attention, which is used in this study. Its form is as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d/h}}\right)V \tag{2}$$

where the $\text{Softmax}(QK^T/\sqrt{d/h})$ can be regarded as the weight of V, indicating the extent to which each element in the time series affects the others. To avoid the query noticing key values appearing after the current element, an upper-left triangle mask is adopted to QK^T.

Finally, the outputs (A) of all the attention heads are merged and then input into the linear transformation layer to obtain the final multihead attention result matrix

$$Z = W_A A + b_A. \tag{3}$$

2) Temporal Transformer Module:

a) Temporal rearrangement of spatiotemporal data: The TIMis designed to capture the temporal correlation of the input Chla spatiotemporal time series, and it optimizes the rearrangement and temporal embedding of the input data based on the transformer encoder (Fig. 1). In general, the transformer can operate on any input sequence $X \in \mathbb{R}^{N\times d}$, where N is the length of the sequence and d is the dimension of each element in the sequence. The main task of the multihead attention mechanism is to extract correlations for the different elements of this sequence. In this study, the Chla spatiotemporal sequence is an atypical form of transformer input that consists of image patches over a period of time ($X_e \in \mathbb{R}^{H\times W\times t}$). To better extract temporal correlations, the original Chla sequences have to be rearranged so that the attention between data at different times can be calculated directly. Thus, the inputted spatiotemporal Chla time series, $X_{1:T} \in \mathbb{R}^{T\times H\times W}$, will first be reshaped into $X^{\tau}_{1:T} \in \mathbb{R}^{T\times(H\times W)}$, where H and W are the image size and T is the length of the input sequence. Then the rearranged data is cast into a d-dimension matrix $X^{\tau}_e \in \mathbb{R}^{T\times d}$ by an embedding layer, allowing the module to further focus on the temporal dependencies. Notably, H and W are equal in this study.

b) Temporal embedding: Theoretically, the computation of the attention mechanism [see (2)] cannot include sequential information about the input time series. However, the dependence of future Chla changes on historical sequences is complex (containing either short or long ranges), i.e., Chla observations may be correlated with observations at more recent times or with observations from a previous period. Therefore, the sequential information of the input time series is critical for the performance of the prediction results. To capture the complex temporal dependencies, we embed

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the sequential information with the input time series in the proposed method.

In this study, each temporal feature in $X_e^{\tau} \in \mathbb{R}^{T\times d}$ is embedded using a sinusoidal position encoding introduced by the original transformer [34]

$$\mathrm{PE}(\mathrm{pos}, 2i) = \sin\left(\mathrm{pos}/10\,000^{\frac{2i}{d_{\mathrm{model}}}}\right) \tag{4}$$

$$\mathrm{PE}(\mathrm{pos}, 2i+1) = \cos\left(\mathrm{pos}/10\,000^{\frac{2i}{d_{\mathrm{model}}}}\right) \tag{5}$$

where pos denotes the position of each element in the time series and i denotes the ith dimension in the input vector. In model construction, the PE is input to the standard embedding layer for positional embedding. Finally, the positional embedding is added to the embedding layer results, making the input to the multiheads attention layer contain temporal positional information.

3) Spatial Transformer Module:

a) Spatial rearrangement of spatiotemporal data: The structure of the STM is similar to that of the TTM but is designed to capture the spatial correlation of the input Chla spatiotemporal time series. To better extract spatial correlations, the original Chla sequences thus have to be rearranged so that the attention between data from different locations can be directly calculated. Therefore, the input $X_{1:T} \in \mathbb{R}^{T\times H\times W}$ is reshaped into $X_{1:T}^{s} \in \mathbb{R}^{(H\times W)\times T}$ in terms of space, where the input sequence length is $H \times W$ and each element has T values. The reshaped data are then cast into a d-dimensional matrix $X_e^{\tau} \in \mathbb{R}^{(H\times W)\times d}$ via embedding layer to facilitate the extraction of spatial dependencies (Fig. 1).

b) Spatial embedding: The natural spatial correlation of satellite images makes the prediction of a target image pixel highly correlated with its neighboring pixels. However, unlike temporal sequential information, spatial correlations are multidirectional, e.g., the closer neighboring pixels are to the target pixel, the higher the similarity is. Thus, a relative position embedding (RPE) approach was used to capture the spatial correlation of the input spatiotemporal sequence.

In this study, the RPE was mapped with a many-to-one function. It allows elements in different positions to share positional embeddings by projecting relative distances to integer sets. According to [40], it is preferable to assign neighboring pixels together so that the spatial information is complete, thereby enhancing the prediction task. Thus, a segmentation function was proposed, which projects the relative distance between two elements into corresponding embeddings and allocates attention accordingly, which is expressed as

$$s(x) = \begin{cases} [x], & |x| \le b \\ \pm\min\left(y, \left[b + \dfrac{In\left(\frac{|x|}{b}\right)}{n\left(\frac{a}{b}\right)}(y-b)\right]\right), & |x| > b \end{cases} \tag{6}$$

where $[x]$ denotes the rounding function. b indicates the splitting point. y represents the output range and a is the curvature of the logarithmic function.

The relative distances $(D(i, j))$ between pixels are measured through Euclidean distances. The result is projected into the corresponding embedding by the proposed mapping function. The RPEs are then expressed as

$$P_{ij} = B_{D^{\bar{x}(i,j)}, D^{\bar{y}(i,j)}}. \tag{7}$$

The vector B, considered a cluster, is trainable and holds the relative-positional weight.

III. Experiments and Analysis

A. Dataset and Study Area

We acquired MODIS-Aqua level-3 daily Chla data with 4 km spatial resolution from the NASA archive (https://oceancolor.gsfc.nasa.gov/l3/) [41]. The experimental data contains 4747 images from January 1, 2012 to December 31, 2022. There exists missing data in the level-3 Chla data caused by clouds, sun glints, and other undesirable conditions, which will affect the implementation of the prediction model. Thus, the data interpolating empirical orthogonal functions (DINEOFs) method was utilized to obtain gap-filled Chla images. Detailed information about DINEOF and related studies on Chla gap-filling can be referred to [9], [38], and [39]. In this study, images with less than 5% valid data in the Chla image sequences were first excluded and then log-transformed for normalization. Cross-validation data were then generated by randomly selecting 3% of valid data and used to determine the best EOF patterns, with a maximum of 30. The final convergence condition for the DINEOF is that the maximum error between the cross-validation data and the reconstruction results reaches the Lanczos convergence threshold of 10^{-8}. The gap-filling accuracy for Chla data in this study is shown in Fig. S1.

The experimental area is located in the eastern China Sea (21°N–43°N and 116°E–127°E), including the Bohai Sea, Yellow Sea, and East China Sea (Fig. 2). There are different types of seas in the experimental area, such as the semi-enclosed Bohai Sea and the semi-open East China Sea, with different annual variations in Chla, e.g., the Bohai Sea has large intraannual fluctuations of Chla, whereas the intraannual variations in the East China Sea are more periodical. Thus, it is possible to explore the performance of the Chla prediction model in different marine environments.

B. Performance Metrics

In this study, three evaluation metrics are used to assess the performance of the proposed prediction models: root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). These metrics are calculated as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i}\left(x_i - \hat{x}_i\right)^2} \tag{8}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|x_i - \hat{x}_i\right| \tag{9}$$

$$\mathrm{MAPE} = \frac{100\%}{n}\sum_{i}\left|\frac{x_i - \hat{x}_i}{x_i}\right| \tag{10}$$

where $\hat{x}$ represents the predicted Chla data and x is predicted by the observed Chla data. n denotes the total number of samples.

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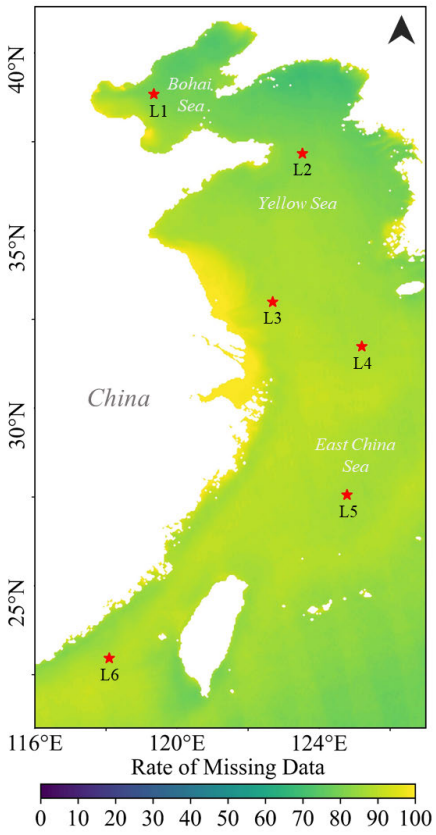


Fig. 2. Location of experimental area and distribution of missing data rates (2012–2022). The stars represent the locations of pixels for temporal recursively prediction.

C. Experimental Settings and Baselines

Chla time series from 2012 to 2022 were utilized to produce the sample dataset. First, the Chla time series was divided into a series of subsets, with each subset separated by one day, and the length of each subset equal to the sum of the length of the historical input data and that of the prediction. Then, samples were extracted cyclically using a fixed window size from the time series images of each subset. In this study, the length of the historical input sequence and the size of the image patch were set to 30 days and 9×9, respectively, to ensure adequate spatiotemporal information input. The output of the sample is a 15-day prediction centered in the image patch. Finally, 400 000 training and 100 000 validation samples were obtained for this study.

The test data were independent of the training data and used the Chla time series from the first 45 days of 2023. The first 30 days were used as input to get the prediction results for the next 15 days. The model accuracy was then analyzed by comparing the prediction results with the real data. Notably, the statistical distribution of the test data was generally consistent with that of the training data, indicating that the test data were sufficiently spatially representative (Fig. S2).

The mean squared error (mse) was used as the loss function. The multihead attention layer for different sub-modules was constructed in a similar way to the standard implementation of the transformer. In this study, it was set up with four heads, each head occupying $1/h$ of the embedding dimension (d). Considering the computation time and memory consumption, the embedding size in the model was set to 128. During the training process, the number of iterations and batch size were set to 100 epochs and 100, respectively. The Adam optimization method was utilized, and training was stopped if the loss stabilized for ten consecutive epochs.

In this study, five DL-based methods including CNN [26], LSTM [28], CNN-LSTM [29], ConvLSTM [17], and Ca-STANet [12], were selected for comparison experiments. Note that the LSTM model received a 1-D sequence from the central pixel of the image patch. All comparative models were trained using identical training and validation datasets, as well as the same training-related hyperparameters (including loss function, training iterations, and batch size) as the proposed model.

D. Experimental Results and Analysis

1) Overall Accuracies of 15-Day Prediction in Different Models: Table I shows the RMSE, MAE, and MAPE results from the different prediction models on the test data. The CNN has the worst performance among the various prediction steps based on daily Chla data since it cannot utilize the temporal correlation of the sequence data. The long-term prediction performance of LSTM was marginally better than that of CNN, which may derive from the fact that the recursive network structure and the gating mechanism of LSTM can better capture the long-term dependencies of time series. Because of the ability to utilize spatiotemporal information, CNN-LSTM and ConvLSTM outperform CNN and LSTM. ConvLSTM outperformed CNN-LSTM in short-term Chla prediction, while CNN-LSTM performed better in medium- and long-term prediction. The possible reason is that the ConvLSTM captures spatial information for memorization through local CNNs, it is prone to lose long-term dependent features when the model layers increase. CNN-LSTM extracts temporal and spatial features separately to preserve the long-term than CONVLSTM on long-term forecasting. Similarly, the Ca-STANet model utilizes the attention mechanism to enhance the modeling of spatiotemporal evolution, resulting in relatively higher prediction accuracy.

Comparatively, the STF_Transformer outperformed all other benchmarks in all metrics, especially in relative long-term (greater than seven days) prediction. Specifically, the RMSE values range from 0.302 to 0.610 mg/m^3, the MAE values range from 0.152 to 0.315 mg/m^3, and the MAPE range from 10.0% to 22.5% for 15-day prediction. Prediction accuracy decreases with increasing prediction dates, a trend common to all models (Fig. 3). Overall, the accuracy of CNN and LSTM decreases obviously as the prediction time increases, and the MAPE on day 15 (52.3%) is much larger than that on day 1 (16.1%). CNN + LSTM, ConvLSTM, and Ca-STANet perform slightly better than CNN and LSTM, but they still have high errors in the mid to late prediction days, with predicted MAPEs of 46.8%, 39.7%, and 37.8% on day 15, respectively. However, our proposed model outperformed the above baseline model. The RMSE for the 15th prediction day

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TABLE I
PERFORMANCES OF DIFFERENT MODELS IN 15-DAY PREDICTING

Models	Metrics	Prediction days					
		1-day	4-day	7-day	10-day	13-day	15-day
CNN	RMSE(mg/m^3)	0.69	0.85	1.08	1.441	1.992	2.821
	MAE(mg/m^3)	0.352	0.436	0.556	0.734	0.998	1.412
	MAPE(%)	16.1	21.9	28.6	39.7	48.8	52.3
LSTM	RMSE(mg/m^3)	0.564	0.682	0.851	1.0968	1.44	1.96
	MAE(mg/m^3)	0.301	0.362	0.455	0.584	0.773	1.06
	MAPE(%)	14.5	19.5	25.9	33.8	41.8	50.7
CNN+LSTM	RMSE(mg/m^3)	0.493	0.581	0.699	0.881	1.119	1.4616
	MAE(mg/m^3)	0.264	0.314	0.382	0.485	0.617	0.803
	MAPE(%)	13.6	17.8	23.8	29.8	37.8	46.8
ConvLSTM	RMSE(mg/m^3)	0.411	0.467	0.554	0.671	0.832	1.063
	MAE(mg/m^3)	0.213	0.245	0.292	0.357	0.442	0.557
	MAPE(%)	12.2	16.9	20.3	25.6	30.4	39.7
Ca-STANet	RMSE(mg/m^3)	0.36	0.402	0.467	0.555	0.677	0.848
	MAE(mg/m^3)	0.198	0.224	0.26	0.311	0.38	0.477
	MAPE(%)	11.9	15.8	19.4	24.5	28.6	37.8
STF_Transformer	RMSE(mg/m^3)	0.302	0.331	0.373	0.426	0.501	0.61
	MAE(mg/m^3)	0.152	0.167	0.19	0.218	0.257	0.315
	MAPE(%)	10.0	13.7	17.3	20.1	21.8	22.5

increased by only 0.308 mg/m^3 compared to the first day, and the MAPE increased by only 12.5%.

Overall, the proposed method outperformed other DL-based methods in relatively long-term (greater than seven days) Chla predictions. The main reason is that the proposed method is more effective in mining multidimensional features of the spatiotemporal evolution of the Chla, which implies that the transformer-based model has promising potential in prediction applications for the ocean.

2) Predictive Performance in Space: Fig. 4 compares the spatial distribution of Chla predictions and observations for six prediction dates between Doy 200 and 215 in 2022. In addition, Fig. 4(m)–(r) shows the spatial distribution of the corresponding RMSEs. Overall, the predictions are in good agreement with the observations, especially for the first seven prediction dates [Fig. 4(g)–(i)], indicating the effectiveness of the proposed model. The RMSE increases with the prediction dates as the prediction error accumulates and propagates from the first date to the last date. Although the accumulation of prediction errors increases the uncertainty in the Chla predictions for the last days, the overall spatial variability of Chla is still reproduced. Spatially, the RMSE is lower in the offshore region than in the nearshore region, mainly due to the high spatial variability and rapid change of Chla in the nearshore region. The RMSE increases with increasing prediction step size in areas where Chla is changing rapidly, while it will be relatively stable in offshore areas. The increase in RMSE is small in the first seven days, and then the increase is relatively raised, which is consistent with the results in Fig. 3. Furthermore, the percentage of pixels with RMSE less than 0.8 (mg/m^3) at the first, 4th, 7th, 10th, 13th, and 15th prediction dates were 97.6%, 97.1%, 96.0%, 93.2%, 91.0%, and 89.3%, respectively. Furthermore, the RMSE spatial distributions of different models were also presented in Fig. 5. Comparatively, the RMSE of the proposed method is lower than the other methods throughout the study area. The results of the model (STF_Transformer, Ca-STANet, and ConvLSTM) that can extract spatiotemporal information are better than the single feature extraction model (LSTM and CNN). Especially in the late prediction period, the proposed method can maintain a low RMSE overall, but the RMSEs of the other methods all show a remarkable increase. These above results suggest that the proposed model has a favorable prediction performance in relative long-term Chla prediction in a large region.

3) Temporal Recursively Predicting: To demonstrate the predictive performance of the proposed model over time, we applied a sliding window to recursively predict the 15-day lead Chla values from 2020 to 2022. Fig. 6 shows the predicted and observed time series of Chla at six locations. Considering the different marine environments, the selected locations are distributed from coastal areas to distant seas (Fig. 2). Generally, the observed Chla time series exhibits two characteristics: 1) the time series in the coastal area shows strong dynamics [Fig. 6(a) and (c)] and 2) the time series in the offshore region shows a more evident periodic variation [Fig. 6(b) and (d)–(f)].

As shown in Fig. 6, the temporal trends of the predicted and observed Chla values at the six different locations are in good agreement, indicating that the proposed model performs well in recursive prediction. In detail, the differences between the predicted and observed time series are slightly higher in the coastal areas [Fig. 6(a) and (c)] than in other locations [Fig. 6(b) and (d)–(f)]. The main reason is that the complex Chla change characteristics along the coastal areas require the model to have a strong dynamic feature extraction capability. The prediction results are more accurate in offshore regions compared to coastal regions. The above results illustrate that the proposed model can effectively utilize

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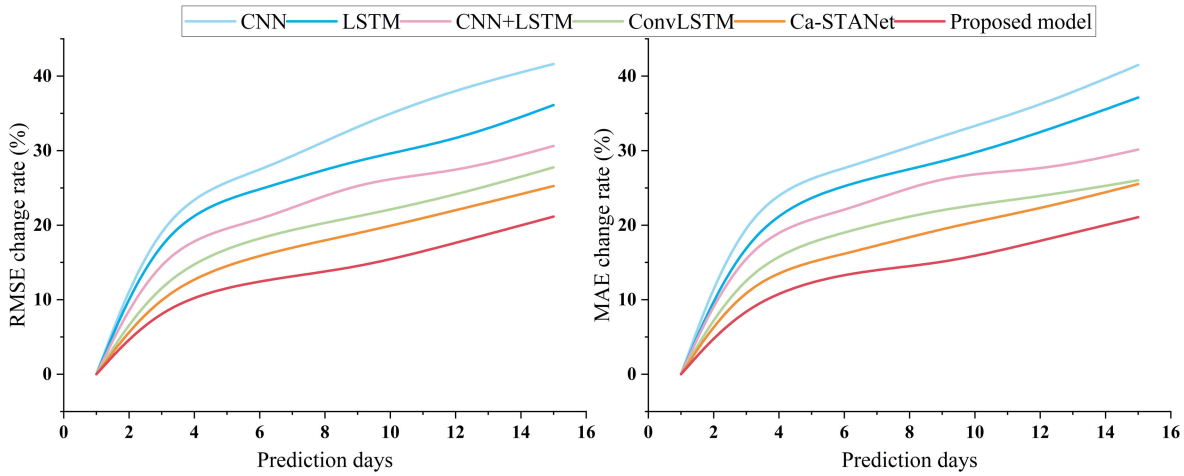


Fig. 3. Proportion of change in accuracy metrics relative to the results of the previous predicted date.

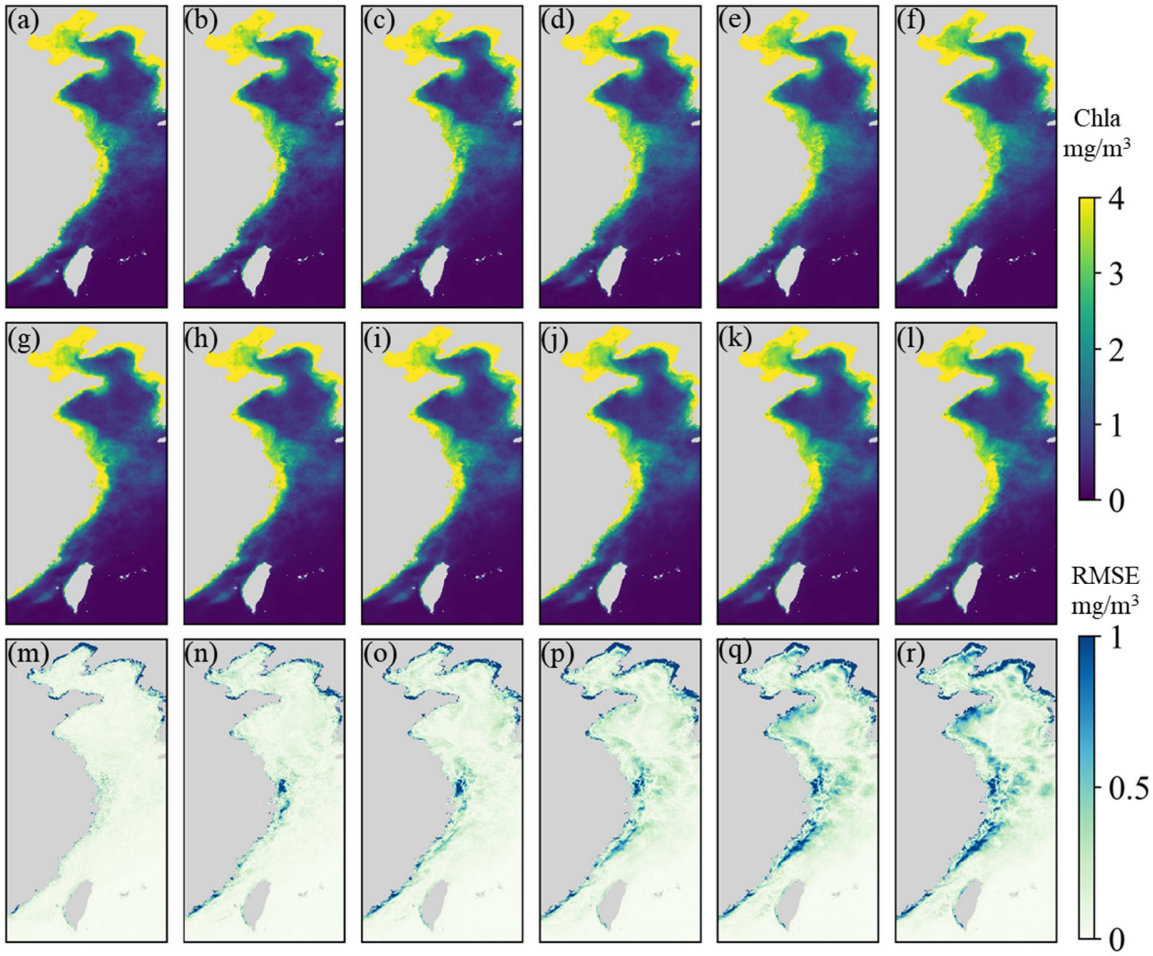


Fig. 4. Spatial distribution of real values (a)–(f), predicted results (g)–(l), and corresponding RMSEs (m)–(r) at different prediction times (1st, 4th, 7th, 10th, 13th, and 15th prediction dates.)

spatiotemporal correlation to predict relatively long-term (15-day) Chla changes in advance.

IV. Discussion

A. Influence of Input Sequence Length

The prediction models are usually affected by the length of the input historical data, especially for ocean Chla with periodic patterns, thus we conducted a comparative experiment to analyze model accuracy by varying the input length. In the experiments, the output length was set to 15, and other settings were maintained. Table II shows the prediction results with varying input lengths. In general, the prediction accuracy increases with the input length but gradually decreases after the input length increases to a certain level. Specifically,

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TABLE II
PERFORMANCES OF THE PROPOSED MODEL WITH DIFFERENT INPUT LENGTHS. THE ACCURACY METRICS ARE THE AVERAGE RESULTS OF THE 15-DAY PREDICTION

Input Length (days)	Prediction Length (days)	RMSE* (mg/m^3)	MAE* (mg/m^3)	MAPE* (%)
10	15	0.467	0.243	23.1
20	15	0.456	2.228	20.3
30	15	0.423	0.216	17.5
40	15	0.420	0.213	17.2
50	15	0.424	0.219	17.7

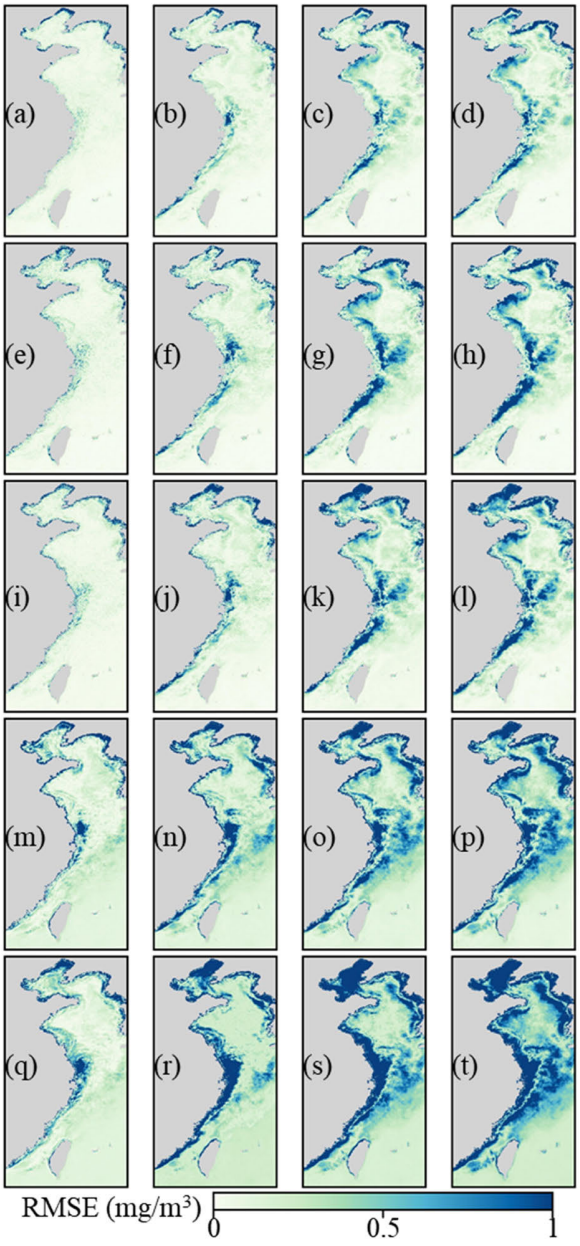


Fig. 5. Spatial distribution of RMSEs at different prediction times (1st, 7th, 13th, and 15th prediction dates) for different models. (a)–(d) STF_Transformer. (e)–(h) Ca-STANet. (i)–(l) ConvLSTM. (m)–(p) LSTM. (q)–(t) CNN.

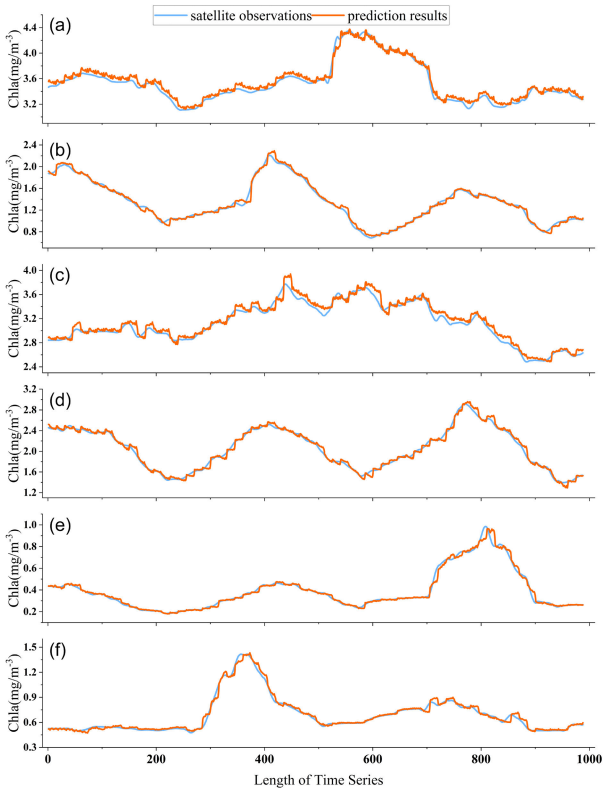


Fig. 6. Predicted and real Chla time series for 15-day lead prediction at different locations. (a) L1, (b) L2, (c) L3, (d) L4, (e) L5, and (f) L6. Note that the time series are filtered by the Savitzky–Golay method to show changes in trends more clearly.

as the input length increases from 10 to 30 days, the accuracy of the model improves, with the RMSE decreasing from 0.512 to 0.484 mg/m^3, the MAE decreasing from 0.296 to 0.273 mg/m^3, and the MAPE decreasing from 24.5% to 18.1%. However, as the input length continues to increase up to 40, the accuracy of the model stabilizes (RMSE only decreases by 0.002 mg/m^3), and as it increases to 50, the accuracy of the model begins to gradually decrease (RMSE increases by 0.004 mg/m^3). Although prediction models based on time-series sequence need to extract change characteristics from historical sequence data to achieve accurate prediction, it is not the case that inputting longer data will result in better accuracy. On the one hand, the longer the input time, the higher the requirement for the model to extract the long-term dependence. On the other hand, the correlation between predicted data and historical data is gradually weakening, causing the input long time series will not appreciably improve

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TABLE III
ABLATION EXPERIMENT RESULTS

Methods	Prediction days	RMSE (mg/m^3)	MAE (mg/m^3)	MAPE (%)
Model_T	1	0.345	0.178	12.4
	9	0.517	0.369	23.4
	15	0.718	0.487	30.4
Model_S	1	0.384	0.197	13.6
	9	0.554	0.397	25.8
	15	0.786	0.521	31.6
Model_ST	1	0.302	0.152	10.0
	9	0.426	0.218	20.1
	15	0.610	0.315	22.5

the model performance. Meanwhile, the accumulation of errors in the long-term series data has the potential to degrade the performance of the model. These results suggest that the input length choice should be close to 30 days to obtain a balance between performance and computational burden in this study.

B. Ablation Studies

To demonstrate the effectiveness of the spatial/TIM, we designed the temporal transformer model (Model_T), the spatial transformer model (Model_S), and the spatiotemporal transformer model (Model_ST) to perform ablation experiments. Among them, the Model_T/Model_S only uses the spatial/TIM to develop the prediction model. As shown in Table III, the RMSE, MAE, and MAPE were higher for the Model_T and Model_S models than for Model_ST. At the 1st, 9th, and 15th predicting dates, the RMSEs increased by 0.043, 0.091, and 0.108 mg/m^3, MAE increased by 0.026, 0.151, and 0.172 mg/m^3 and MAPE increased by 2.4%, 3.4%, and 7.9% for Model_T. The accuracy of Model_S was much lower, with RMSE increasing by 0.082, 0.128, and 0.176 mg/m^3, MAE increased by 0.045, 0.179, and 0.206 mg/m^3, and MAPE increasing by 3.6%, 5.7%, and 9.1%. Overall, the accuracy of Model_T is generally higher than that of Model_S, while the inclusion of spatiotemporal information markedly improves the performance of Model_ST. This is attributed to the fact that temporally dependent features play a dominant role in spatiotemporal sequence prediction tasks. Additionally, the accuracy improvement brought by spatial information suggests that the effective extraction of spatially dependent features can improve the prediction model performance when the predictor variables are spatiotemporally correlated. The above results indicate that the combination of temporal and STMs can augment the extraction capability of spatiotemporal information in the proposed model.

C. Advantages and Limitations

In this study, the objective was to accurately predict relatively long-term Chla (15 steps ahead) over a large area based on a deep learning approach. The main challenge is to accurately capture variable dynamic (long and short) dependencies across space and time. The proposed model (STF-Transformer) better solves this problem by capturing the complex spatiotemporal dependencies in the data through the dynamic receptive field of multiple-head attention.

Compared to existing DL-based methods, this study contributes a method that realizes Chla time series prediction using only the attention mechanism. Experimental results show that the performances of the STF-Transformer model outperform other existing prediction models, which is attributed to the fact that the multihead attention mechanism is more effective in capturing complex dependencies across spatiotemporal dimensions. Our model has two advantages when confronted with the task of long-term Chla prediction over large areas: 1) the transformer-based network can capture the spatiotemporal dynamic correlations of the input data at once through the attentional mechanism, whereas the current learning-based methods are mainly based on RNNs and convolutional networks to extract them, which suffers from the long-term dependence vanishing and insufficient receptive fields and 2) based on the aggregated information extracted from the temporally and spatially transformed modes, the STF-Transformer model can predict Chla dynamics for different scenarios over a large area, such as high dynamics variations in the inshore sea and periodic variations in the offshore sea. These advantages of the proposed model will help to better support marine ecosystem health monitoring and provide early warning of algal blooms.

Although the proposed model has obtained favorable performance in Chla prediction, there are still some limitations. One concern is that the multihead self-attention and feed-forward network models, which are the core of the transformer, need to learn a large number of parameters, which imposes a training burden on model development. In this study, we limit the spatial window size of the input data and the dimensionality of the embedding layer to achieve a balance between training burden and accuracy. Thus, a lightweight design of the model is required when trying to extract features from longer and larger spatiotemporal series data. Another concern is that the learned parameters of the proposed model are related to the spatial region, it is necessary to add training data to improve the applicability when applied in other regions. Additionally, the reconstruction accuracy of missing data in the Chla time series will affect the quality of the training samples, hence an accurate gap-filling method is necessary.

V. CONCLUSION

Predicting Chla can provide support for ocean environment monitoring, fishery management, and early warning of algae

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blooms. This study developed a spatiotemporal feature fusion model for Chla prediction based on a transformer framework. This model consisted of a TIM and an STM, which were used to extract long-term and short-term temporal characteristics and spatial dependencies from historical Chla image sequences. The evaluation results showed that the proposed model achieved favorable performance in Chla prediction, with RMSEs, MAE, and MAPE less than 0.61 mg/m^3, 0.315 mg/m^3, and 22.5% for 15 days prediction in a large area (including Bohai, Yellow, and East China Sea), respectively. Compared to other models (including CNN, LSTM, CNN-LSTM, ConvLSTM, and Ca-STANet), the proposed model achieved the overall best prediction accuracy, and its accuracy degradation was smaller as the prediction period increased, suggesting that more efficient spatiotemporal feature extraction is crucial for relatively long-term Chla prediction. In addition, the predicted Chla showed good agreement with the actual observations in both space and time, further demonstrating the reliability of the proposed model. This work presents a Chla prediction model that allows for 15-day (or 15-step) large-area predictions, providing innovative insights into the prediction of changes in the marine environment over a relatively long term in a large area.

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